

Literature Status: Sharp CHS-to-Operator Lower Constant

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Status. literature_already_answered.

Source question. Bouthat, Chávez, and Garcia, *Hunter’s positivity theorem and random vector norms*, arXiv:2403.10314, Problem 5 on PDF page 59, asks: “Find a sharp lower bound in Theorem 10.5.” For even d , that theorem concerns Hermitian matrices and

$$\|A\|_d^d = h_d(\lambda_1(A), \dots, \lambda_n(A)), \quad \|A\|_{\text{op}} = \max_i |\lambda_i(A)|.$$

Supporting answer. Brazitikos and Pandis, *Sharp inequalities for symmetric polynomials, Hunter’s conjecture, and moments of exponential random variables*, arXiv:2512.12254, Section 6, Theorem 6.1, Lemmas 6.2–6.3, and Corollary 6.4 (starting on PDF page 37), compute the exact minimum of h_{2k} on the ℓ_∞ -sphere. Their introduction (PDF page 5) explicitly states that Section 6 answers the sharp matrix-norm lower-constant question.

Identification. Let $d = 2k$, $n \geq 2$, and let $\tau_{n,k} \in (-1, 0)$ be the unique root of

$$h_{2k-1}(\underbrace{\tau_{n,k}, \dots, \tau_{n,k}}_{n \text{ copies}}, 1) = 0.$$

The supporting result gives

$$\min_{\|x\|_\infty=1} h_{2k}(x) = \binom{n+2k-1}{2k} |\tau_{n,k}|^{2k}.$$

Therefore the sharp constant sought in Problem 5 is

$$C_{n,2k} = \left(\binom{n+2k-1}{2k} \right)^{1/(2k)} |\tau_{n,k}|, \quad C_{n,2k} \|A\|_{\text{op}} \leq \|A\|_{2k}.$$

Equality is realized, up to scale, global sign, and permutation, by a Hermitian matrix whose eigenvalue vector is $(\tau_{n,k}, \dots, \tau_{n,k}, 1)$, with $n-1$ copies of $\tau_{n,k}$.

Indexing note. The displayed statement of Theorem 6.1 writes $n-1$ copies in the odd-degree root equation. Lemmas 6.2–6.3, Corollary 6.4, and the stationarity calculation use the correct equation with n copies; the extremizing vector itself has $n-1$ copies. For instance, $n=2$, $k=1$ requires the root of $2t+1$.

Scope. This resolves Problem 5 only and is recorded as literature retrieval, not as a new result of the run. The other problems in the source paper are not addressed here.

References

- [1] L. Bouthat, A. Chávez, and S. R. Garcia, *Hunter’s positivity theorem and random vector norms*, arXiv:2403.10314 (2024), <https://arxiv.org/abs/2403.10314>.
- [2] S. Brazitikos and C. Pandis, *Sharp inequalities for symmetric polynomials, Hunter’s conjecture, and moments of exponential random variables*, arXiv:2512.12254 (2025), <https://arxiv.org/abs/2512.12254>.